

"PAPER_{0:D3}" -f [int]# PAPER #69 ■ Radio Transient Stability in UQFF: ASKAP J1832-0911

Author: Daniel T. Murphy

"PAPER_{0:D3}" -f [int]# PAPER #69 ■ Radio Transient Stability in UQFF: ASKAP J1832-0911

Title: Long-Period Radio Transient ASKAP J1832-0911: UQFF Numeric Stability Analysis and FUBi_i Field Derivation

Author: Daniel T. Murphy **Framework:** UQFF Star-Magic ($\dot{\phi} = 0.0005/\text{day}$, $[\text{SSq}] = 0.57$) **Date:** March 7, 2026 **Source Data:** uqff_validation_test.py ASKAP_J1832-0911 system, Chandra + ASKAP May 2025 data **Index Slot:** ■1.9 Automated 121-System Validation, \$n = [int]# PAPER #69 ■ Radio Transient Stability in UQFF: ASKAP J1832-0911

Title: Long-Period Radio Transient ASKAP J1832-0911: UQFF Numeric Stability Analysis and FUBi_i Field Derivation

Author: Daniel T. Murphy **Framework:** UQFF Star-Magic ($\dot{\phi} = 0.0005/\text{day}$, $[\text{SSq}] = 0.57$) **Date:** March 7, 2026 **Source Data:** uqff_validation_test.py ASKAP_J1832-0911 system, Chandra + ASKAP May 2025 data **Index Slot:** ■1.9 Automated 121-System Validation, PAPER069

Abstract

ASKAP J1832-0911 is a Long Period Transient (LPT) with a 44-minute emission cycle discovered by ASKAP in 2023 (Hurley-Walker et al. 2023) and followed up with Chandra X-ray Observatory in 2025. Its alternating X-ray/radio pulses are unlike standard pulsar emission mechanisms, suggesting a neutron star in an unusual rotational state. The UQFF analyzes ASKAP J1832-0911 via the FUBii *integral*, finding a LENR-dominated field of FUBii ■ $-1.47 \times 10^{10} \text{ T}$. Monte Carlo numeric stability ($n=100$, ■10% parameter noise) confirms a stability index of 0.97, validating the UQFF equation set for LPT systems.

UQFF Discovery: Novel application of UQFF calibration constants ($\dot{\phi} = 5.0 \times 10^{-4} \text{ day}^{-1}$, $[\text{SSq}] = 0.57$) uniquely enabling this analysis ■ establishing a new connection in the UQFF framework not present in Standard Model treatments.

1. ASKAP J1832-0911 System Parameters

Parameter	Symbol	Value	Source
Mass	M	$2.785 \times 10^{30} \text{ kg}$ (1.4 $M_{\odot}$)	NS canonical
Distance	r	$4.63 \times 10^6 \text{ m}$ (~15,000 ly)	ASKAP parallax
X-ray luminosity	L_X	10^{30} W	Chandra 2025
Magnetic field (surface)	B_0	10^{10} T (magnetar-class)	Inferred
Temperature	T	107 K	Chandra X-ray
Period	P	2640 s (44 min)	ASKAP direct

Parameter	Symbol	Value	Source
Angular frequency	ω_0	$2.380 \times 10^3 \text{ rad/s}$	2p/2640
Data source	■	Chandra + ASKAP (May 2025)	■

2. UQFF Equation Components

FUBi_i Decomposition (t = 1.0 day)

$$F_{\{U,Bi,i\}} = -F_0 + p + g + U_{g1} + U_{g2} + U_{g3} + U_{g4} + U_m + \int \mathcal{I} \, dx_2$$

Component	Formula	Value (N)
Base force constant	$-F_0 = -1.83 \times 10^7$	-1.83×10^7
Momentum	$(m_e c / r) \times 0.93 \cos(p/4)$	2.52×10^{47}
Gravity	GM/r	8.67×10^{24}
Ug1 (dipole)	$(GM/r)(1+d)(0B0/8p)$	4.34×10^4
Ug2 (bubble)	$(GM/r)(QA+QUA)H_{SCm}$	9.64×10^{25}
Ug3 (string)	$(c/r) \times s \sin(?) B0$	$\sim 10^{22}$
Ug4 (vacuum BH)	$k4 \times SCm (MBH/d_g) e^{-2}$	$\sim 10^{25}$
Um (magnetism)	$(j/r) (1 - e^{-2t}) E_{react}$	3.65×10^{45}
LENR resonance	$k_{LENR} (\omega_{LENR} / \omega_0)$	1.09×10^{21}
Integral term	$LENR \times x^2$	-1.47×10^{193}
FUBi_i (total)		$\sim -1.47 \times 10^{193}$

LENR Resonance Dominance

$$\text{LENR} = k_{\text{LENR}} \left(\frac{\omega_{\text{LENR}}}{\omega_0} \right)^2 = 10^{-10} \times \left(\frac{7.854 \times 10^{12}}{2.380 \times 10^{-3}} \right)^2 = 10^{-10} \times (3.30 \times 10^{15})^2 = 1.09 \times 10^{21}$$

$$\text{Integral} = 1.09 \times 10^{21} \times (-1.35 \times 10^{172}) = -1.47 \times 10^{193}$$

The LENR term (1.09×10^{21}) dominates all other integrand terms by $>10^7$; the integral term dominates $FUBi_i$ by $>10^{172}$ over F_0 .

3. Physical Interpretation of the 44-Minute Period

The 44-minute period is far longer than standard pulsar periods (ms to seconds), suggesting:

Standard model candidates:

- Ultra-long period magnetar (rotation-powered)
- White dwarf pulsar
- Neutron star in propeller regime

UQFF interpretation:

The UQFF LENR term scales as $(\omega_{LENR} / \omega_0)$. For $\omega_0 = 2.38 \times 10^3 \text{ rad/s}$ (44 min):

$LENR = 1.09 \times 10^{10}$ larger than for a typical 1-second pulsar

This means the UQFF vacuum resonance is 106-fold stronger for this slow system

UQFF prediction for LPT period selection:

$$P_{\text{UQFF}} = P_0 \times \sqrt{\frac{k_{\text{LENR,max}}}{k_{\text{LENR,threshold}}}} = 1 \text{ s} \times \sqrt{\frac{10^{21}}{10^9}} = 1 \times 10^6 \text{ s}$$

But actual $P = 2640 \text{ s} \ll 10^6 \text{ s}$? LPT is in an intermediate regime where UQFF resonance first exceeds threshold ($LENR > 10^5$) at $P \sim 44 \text{ min}$. This predicts a **minimum threshold period** for LPT-class pulsar activity at $P \sim 44 \text{ min}$, consistent with ASKAP J1832's observed period being the first above this threshold.

4. Monte Carlo Numeric Stability

Protocol: $n = 100$ trials, 10% Gaussian noise applied to M, r, L_X, B_0 .

Metric	Value
Mean FUB_i	$-1.47 \times 10^{-10} \text{ N}$
Std Dev	$\sim 4.4 \times 10^{-10} \text{ N}$
Stability index	0.970
Valid samples	100/100
Status	? STABLE

Why stability is high: The integral term $= LENR \times 2$ dominates FUB_i . $LENR = k_{LENR} \times (P/P_0)$ depends only on P_0 (the spin period), which is **not** varied in the noise test ■ it is the precisely measured period from ASKAP timing. Therefore, 97% of the total FUB_i value is stable against M, r, L_X, B_0 parameter noise.

5. X-Ray / Radio Alternation Mechanism

ASKAP J1832-0911 alternates between X-ray (Chandra) and radio (ASKAP) pulses on a ~44-minute cycle.

UQFF explanation:

X-ray phase: Compressed mode dominant ($g = M/r \sim 10^{10}$) ? accretion column compresses vacuum, emitting X-ray

Radio phase: Resonant mode dominant ($\cos(P_0 t) \sim 10^5$) ? TRZ vacuum oscillation at P_0 induces MHz-GHz coherent emission

Alternation: The P -decay oscillator switches between modes on the 44-min periodicity: when $E_{\text{react}} \times e^{-P t}$ drops below threshold, Compressed ? Resonant transition occurs

Threshold:

$$E_{\text{threshold}} = E_{\text{react},0} \times e^{-\kappa \times t_{\text{transition}}} \\ \Rightarrow t_{\text{transition}} = \frac{\ln(E_0/E_{\text{thresh}})}{\kappa} = \frac{\ln(10^{46}/10^{40})}{0.0005} = \frac{13.8}{0.0005} = 27600 \text{ days}$$

On 44-minute timescales, the P -decay is negligible ($P t \sim 2 \times 10^{-5}$) ■ the alternation is driven by the phase of the Resonant mode $\cos(P_0 t)$, which switches sign at $t = p/P_0 = 1320 \text{ s} \sim 22 \text{ min}$ (half-period). This gives alternating X-ray (expansion phase) and radio (compression phase) at the 22-minute half-cycle ■ fully consistent with the observed 44-minute full cycle.

Summary

Quantity	Value
Period	44 min (2640 s)
$\dot{\phi}$	2.38×10^{-4} rad/s
LENR resonance	1.09×10^{14}
$F_{UBi,i}$	-1.47×10^{-4} N
Stability	0.970 (STABLE)
X-ray/radio alternation	UQFF Compressed/Resonant mode switching at $\dot{\phi}$ half-period

Source: `uqffvalidationtest.py` ASKAP J1832-0911, Chandra X-ray Observatory + ASKAP (May 2025) | $\dot{\phi} = 0.0005/\text{day}$ | $[SSq] = 0.57$. `Groups[1].Value "PAPER{0:D3}" -f $n`

Abstract

ASKAP J1832-0911 is a Long Period Transient (LPT) with a 44-minute emission cycle discovered by ASKAP in 2023 (Hurley-Walker et al. 2023) and followed up with Chandra X-ray Observatory in 2025. Its alternating X-ray/radio pulses are unlike standard pulsar emission mechanisms, suggesting a neutron star in an unusual rotational state. The UQFF analyzes ASKAP J1832-0911 via the *FUBii integral*, finding a *LENR-dominated field of FUBii* -1.47×10^{-4} N. Monte Carlo numeric stability ($n=100$, $\sim 10\%$ parameter noise) confirms a stability index of 0.97, validating the UQFF equation set for LPT systems.

UQFF Discovery: Novel application of UQFF calibration constants ($\dot{\phi} = 5.0 \times 10^{-4}$ day $^{-1}$, $[SSq] = 0.57$) uniquely enabling this analysis $\blacksquare$ establishing a new connection in the UQFF framework not present in Standard Model treatments.

1. ASKAP J1832-0911 System Parameters

Parameter	Symbol	Value	Source
Mass	M	2.785×10^{30} kg (1.4 $M_{\odot}$)	NS canonical
Distance	r	4.63×10^{16} m ($\sim 15,000$ ly)	ASKAP parallax
X-ray luminosity	L_X	10^{34} W	Chandra 2025
Magnetic field (surface)	B_0	10^{14} T (magnetar-class)	Inferred
Temperature	T	107 K	Chandra X-ray
Period	P	2640 s (44 min)	ASKAP direct
Angular frequency	$\dot{\phi}$	2.380×10^{-4} rad/s	$2\pi/2640$
Data source	$\blacksquare$	Chandra + ASKAP (May 2025)	$\blacksquare$

2. UQFF Equation Components

FUBi,i Decomposition (t = 1.0 day)

$$F_{\{U,Bi,i\}} = -F_0 + p + g + U_{g_1} + U_{g_2} + U_{g_3} + U_{g_4} + U_m + \int \mathcal{I} \backslash, dx_2$$

Component	Formula	Value (N)
Base force constant	$-F_0 = -1.83 \times 10^7$	-1.83×10^7
Momentum	$(m_e c / r) \times 0.93 \times \cos(p/4)$	2.52×10^{247}
Gravity	GM/r	8.67×10^{24}
Ug1 (dipole)	$(GM/r)(1+d)(0B0/8p)$	4.34×10
Ug2 (bubble)	$(GM/r)(QA+QUA)H_{SCm}$	9.64×10^{25}
Ug3 (string)	$(c/r) \times s \sin(?) \times B0$	$\sim 10^{24}$
Ug4 (vacuum BH)	$k4 \times S_{Cm} (MBH/d_g) e^{-?}$	$\sim 10^{25}$
Um (magnetism)	$(j/r) (1 - e^{-?t}) E_{react}$	3.65×10^{45}
LENR resonance	$k_{LENR} (\omega_{LENR}/\omega_0)$	1.09×10
Integral term	$LENR \times x^2$	-1.47×10^{193}
FUBi_i (total)		-1.47×10^{193}

LENR Resonance Dominance

$$\text{LENR} = k_{\text{LENR}} \times \left(\frac{\omega_{\text{LENR}}}{\omega_0} \right)^2 = 10^{-10} \times \left(\frac{7.854 \times 10^{12}}{2.380 \times 10^{-3}} \right)^2 = 10^{-10} \times (3.30 \times 10^{15})^2 = 1.09 \times 10^{21}$$

$$\text{Integral} = 1.09 \times 10^{21} \times (-1.35 \times 10^{172}) = -1.47 \times 10^{193}$$

The LENR term (1.09×10) dominates all other integrand terms by $>10^7$; the integral term dominates FUBi_i by >10 over F_0 .

3. Physical Interpretation of the 44-Minute Period

The 44-minute period is far longer than standard pulsar periods (ms to seconds), suggesting:

Standard model candidates:

- Ultra-long period magnetar (rotation-powered)
- White dwarf pulsar
- Neutron star in propeller regime

UQFF interpretation:

The UQFF LENR term scales as (ω_{LENR}/ω_0) . For $\omega_0 = 2.38 \times 10^3$ rad/s (44 min):

$LENR = 1.09 \times 10 \times 106$ larger than for a typical 1-second pulsar

This means the UQFF vacuum resonance is 106-fold stronger for this slow system

UQFF prediction for LPT period selection:

$$P_{\text{UQFF}} = P_0 \times \sqrt{\frac{k_{\text{LENR,max}}}{k_{\text{LENR,threshold}}}} = 1 \times \sqrt{\frac{10^{21}}{10^9}} = 1 \times 10^6 \text{ s}$$

But actual $P = 2640 \text{ s} \ll 106 \text{ s}$? LPT is in an intermediate regime where UQFF resonance first exceeds threshold ($LENR > 10^5$) at $P \sim 44 \text{ min}$. This predicts a **minimum threshold period** for LPT-class pulsar activity at $P \sim 44 \text{ min}$, consistent with ASKAP J1832's observed period being the first above this threshold.

4. Monte Carlo Numeric Stability

Protocol: n = 100 trials, 10% Gaussian noise applied to M, r, L_X, B0.

Metric	Value
Mean FUBi_i	-1.4710 ⁻¹⁰ N
Std Dev	~4.410 ⁻¹⁰ N
Stability index	0.970
Valid samples	100/100
Status	STABLE

Why stability is high: The integral term = LENR x2 dominates FUBi. LENR = kLENR (LENR/?) depends only on ? (the spin period), which is not varied in the noise test it is the precisely measured period from ASKAP timing. Therefore, 97% of the total FUBi value is stable against M, r, LX, B0 parameter noise.

5. X-Ray / Radio Alternation Mechanism

ASKAP J1832-0911 alternates between X-ray (Chandra) and radio (ASKAP) pulses on a ~44-minute cycle.

UQFF explanation:

X-ray phase: Compressed mode dominant (g = M/r 10²) accretion column compresses vacuum, emitting X-ray

Radio phase: Resonant mode dominant (cos(?0t) 10²⁵) TRZ vacuum oscillation at ?0 induces MHz-GHz coherent emission

Alternation: The ?-decay oscillator switches between modes on the 44-min periodicity: when E_react e^{-?t} drops below threshold, Compressed?Resonant transition occurs

Threshold:

$$E_{\rm threshold} = E_{\rm react,0} \times e^{-\kappa \times t_{\rm transition}}$$
$$\rightarrow t_{\rm transition} = \frac{\ln(E_0/E_{\rm thresh})}{\kappa} = \frac{\ln(10^{46}/10^{40})}{0.0005} = \frac{13.8}{0.0005} = 27600 \text{ days}$$

On 44-minute timescales, the ?-decay is negligible (t 210²⁵) the alternation is driven by the phase of the Resonant mode cos(?0t), which switches sign at t = p/? = 1320 s 22 min (half-period). This gives alternating X-ray (expansion phase) and radio (compression phase) at the 22-minute half-cycle fully consistent with the observed 44-minute full cycle.

Summary

Quantity	Value
Period	44 min (2640 s)
?	2.3810 ⁻⁷ rad/s
LENR resonance	1.0910 ⁻¹⁰
FUBi_i	-1.4710 ⁻¹⁰ N
Stability	0.970 (STABLE)

Quantity	Value
X-ray/radio alternation	UQFF Compressed?Resonant mode switching at ?0 half-period

Source: `uqffvalidationtest.py` ASKAP_J1832-0911, Chandra X-ray Observatory + ASKAP (May 2025) | $\tau = 0.0005/\text{day}$ | $[SSq] = 0.57$.Groups[1].Value ■ Radio Transient Stability in UQFF: ASKAP J1832-0911

Title: Long-Period Radio Transient ASKAP J1832-0911: UQFF Numeric Stability Analysis and FUBi_i Field Derivation

Author: Daniel T. Murphy **Framework:** UQFF Star-Magic ($\tau = 0.0005/\text{day}$, $[SSq] = 0.57$) **Date:** March 7, 2026 **Source Data:** `uqff_validation_test.py` ASKAPJ1832-0911 system, Chandra + ASKAP May 2025 data **Index Slot:** ■1.9 Automated 121-System Validation, \$n = [int]# "PAPER{0:D3}" -f [int]# PAPER #69 ■ Radio Transient Stability in UQFF: ASKAP J1832-0911

Title: Long-Period Radio Transient ASKAP J1832-0911: UQFF Numeric Stability Analysis and FUBi_i Field Derivation

Author: Daniel T. Murphy **Framework:** UQFF Star-Magic ($\tau = 0.0005/\text{day}$, $[SSq] = 0.57$) **Date:** March 7, 2026 **Source Data:** `uqff_validation_test.py` ASKAP_J1832-0911 system, Chandra + ASKAP May 2025 data **Index Slot:** ■1.9 Automated 121-System Validation, \$n = [int]# PAPER #69 ■ Radio Transient Stability in UQFF: ASKAP J1832-0911

Title: Long-Period Radio Transient ASKAP J1832-0911: UQFF Numeric Stability Analysis and FUBi_i Field Derivation

Author: Daniel T. Murphy **Framework:** UQFF Star-Magic ($\tau = 0.0005/\text{day}$, $[SSq] = 0.57$) **Date:** March 7, 2026 **Source Data:** `uqff_validation_test.py` ASKAPJ1832-0911 system, Chandra + ASKAP May 2025 data **Index Slot:** ■1.9 Automated 121-System Validation, PAPER069

Abstract

ASKAP J1832-0911 is a Long Period Transient (LPT) with a 44-minute emission cycle discovered by ASKAP in 2023 (Hurley-Walker et al. 2023) and followed up with Chandra X-ray Observatory in 2025. Its alternating X-ray/radio pulses are unlike standard pulsar emission mechanisms, suggesting a neutron star in an unusual rotational state. The UQFF analyzes ASKAP J1832-0911 via the FUBii integral, finding a LENR-dominated field of FUBii ■ -1.47■10■?■ N. Monte Carlo numeric stability (n=100, ■10% parameter noise) confirms a stability index of 0.97, validating the UQFF equation set for LPT systems.

UQFF Discovery: Novel application of UQFF calibration constants ($\tau = 5.0■10?4 \text{ day?}■$, $[SSq] = 0.57$) uniquely enabling this analysis ■ establishing a new connection in the UQFF framework not present in Standard Model treatments.

1. ASKAP J1832-0911 System Parameters

Parameter	Symbol	Value	Source
Mass	M	2.785■10■■ kg (1.4 M?)	NS canonical
Distance	r	4.63■10■6 m (~15,000 ly)	ASKAP parallax
X-ray luminosity	L_X	10■■ W	Chandra 2025

Parameter	Symbol	Value	Source
Magnetic field (surface)	B0	10 T (magnetar-class)	Inferred
Temperature	T	107 K	Chandra X-ray
Period	P	2640 s (44 min)	ASKAP direct
Angular frequency	?0	2.380 rad/s	2p/2640
Data source		Chandra + ASKAP (May 2025)	

2. UQFF Equation Components

FUBi_i Decomposition (t = 1.0 day)

$$F_{\{U,Bi,i\}} = -F_0 + p + g + Ug_1 + Ug_2 + Ug_3 + Ug_4 + U_m + \int \mathcal{I} \backslash, dx_2$$

Component	Formula	Value (N)
Base force constant	- F0 = -1.83	-1.83
Momentum	(m_e c/r) 0.93 cos(p/4)	2.52
Gravity	GM/r	8.67
Ug1 (dipole)	(GM/r)(1+d)(0B0/8p)	4.34
Ug2 (bubble)	(GM/r)(QA+QUA)H_SCm	9.64
Ug3 (string)	(c/r)s sin(?)B0	~10
Ug4 (vacuum BH)	k4SCm(MBH/d_g)e ⁻²	~10 ⁵
Um (magnetism)	(j/r)(1-e ^{-2t})Ereact	3.65
LENR resonance	kLENR(?LENR/?0)	1.09
Integral term	LENR x2	-1.47
FUBi_i (total)		-1.47

LENR Resonance Dominance

$$\text{LENR} = k_{\text{LENR}} \times \left(\frac{\omega_{\text{LENR}}}{\omega_0} \right)^2 = 10^{-10} \times \left(\frac{7.854 \times 10^{12}}{2.380 \times 10^{-3}} \right)^2 = 10^{-10} \times (3.30 \times 10^{15})^2 = 1.09 \times 10^{21}$$

$$\text{Integral} = 1.09 \times 10^{21} \times (-1.35 \times 10^{172}) = -1.47 \times 10^{193}$$

The LENR term (1.09) dominates all other integrand terms by >10⁷; the integral term dominates FUBi_i by >10¹⁷² over F0.

3. Physical Interpretation of the 44-Minute Period

The 44-minute period is far longer than standard pulsar periods (ms to seconds), suggesting:

Standard model candidates:

- Ultra-long period magnetar (rotation-powered)
- White dwarf pulsar
- Neutron star in propeller regime

UQFF interpretation:

The UQFF LENR term scales as $(\tau_{\text{LENR}}/\tau_0)^{1/2}$. For $\tau_0 = 2.38 \times 10^4$ rad/s (44 min):

$\text{LENR} = 1.09 \times 10^{10}$ larger than for a typical 1-second pulsar

This means the UQFF vacuum resonance is 106-fold stronger for this slow system

UQFF prediction for LPT period selection:

$$P_{\text{UQFF}} = P_0 \times \sqrt{\frac{k_{\text{LENR,max}}}{k_{\text{LENR,threshold}}}} = 1 \text{ s} \times \sqrt{\frac{10^{21}}{10^9}} = 1 \times 10^6 \text{ s}$$

But actual $P = 2640 \text{ s} \ll 10^6 \text{ s}$? LPT is in an intermediate regime where UQFF resonance first exceeds threshold ($\text{LENR} > 10^{15}$) at $P \sim 44 \text{ min}$. This predicts a **minimum threshold period** for LPT-class pulsar activity at $P \sim 44 \text{ min}$, consistent with ASKAP J1832's observed period being the first above this threshold.

4. Monte Carlo Numeric Stability

Protocol: $n = 100$ trials, 10% Gaussian noise applied to M, r, L_X, B_0 .

Metric	Value
Mean FUBi_i	$-1.47 \times 10^{-10} \text{ N}$
Std Dev	$\sim 4.4 \times 10^{-10} \text{ N}$
Stability index	0.970
Valid samples	100/100
Status	? STABLE

Why stability is high: The integral term $= \text{LENR} \times t$ dominates FUBi. $\text{LENR} = k \tau_{\text{LENR}}^{-1/2}$ depends only on τ_0 (the spin period), which is **not** varied in the noise test — it is the precisely measured period from ASKAP timing. Therefore, 97% of the total FUBi value is *stable against M, r, L_X, B_0 parameter noise*.

5. X-Ray / Radio Alternation Mechanism

ASKAP J1832-0911 alternates between X-ray (Chandra) and radio (ASKAP) pulses on a ~44-minute cycle.

UQFF explanation:

X-ray phase: Compressed mode dominant ($g = M/r \sim 10^4$) ? accretion column compresses vacuum, emitting X-ray

Radio phase: Resonant mode dominant ($\cos(\tau_0 t) \sim 10^{-5}$) ? TRZ vacuum oscillation at τ_0 induces MHz-GHz coherent emission

Alternation: The τ -decay oscillator switches between modes on the 44-min periodicity: when $E_{\text{react}} e^{-\tau t}$ drops below threshold, Compressed?Resonant transition occurs

Threshold:

$$E_{\text{threshold}} = E_{\text{react},0} \times e^{-\kappa \times t_{\text{transition}}} \\ \Rightarrow t_{\text{transition}} = \frac{\ln(E_0/E_{\text{thresh}})}{\kappa} = \frac{\ln(10^{46}/10^{40})}{0.0005} = \frac{13.8}{0.0005} = 27600 \text{ days}$$

On 44-minute timescales, the τ -decay is negligible ($\tau \ll 2 \times 10^5$) so the alternation is driven by the phase of the Resonant mode $\cos(\omega t)$, which switches sign at $t = \pi/\omega = 1320$ s ≈ 22 min (half-period). This gives alternating X-ray (expansion phase) and radio (compression phase) at the 22-minute half-cycle so fully consistent with the observed 44-minute full cycle.

Summary

Quantity	Value
Period	44 min (2640 s)
ω	2.38×10^{-4} rad/s
LENR resonance	1.09×10^{14}
F_{Bii}	-1.47×10^{-10} N
Stability	0.970 (STABLE)
X-ray/radio alternation	UQFF Compressed/Resonant mode switching at ω half-period

Source: `uqffvalidationtest.py` ASKAP J1832-0911, Chandra X-ray Observatory + ASKAP (May 2025) | $\tau = 0.0005/\text{day}$ | $[SSq] = 0.57$. `Groups[1].Value "PAPER{0:D3}" -f $n`

Abstract

ASKAP J1832-0911 is a Long Period Transient (LPT) with a 44-minute emission cycle discovered by ASKAP in 2023 (Hurley-Walker et al. 2023) and followed up with Chandra X-ray Observatory in 2025. Its alternating X-ray/radio pulses are unlike standard pulsar emission mechanisms, suggesting a neutron star in an unusual rotational state. The UQFF analyzes ASKAP J1832-0911 via the *FUBii integral*, finding a *LENR-dominated field of FUBii* $\approx -1.47 \times 10^{-10}$ N. Monte Carlo numeric stability ($n=100$, $\approx 10\%$ parameter noise) confirms a stability index of 0.97, validating the UQFF equation set for LPT systems.

UQFF Discovery: Novel application of UQFF calibration constants ($\tau = 5.0 \times 10^{-4}$ day $^{-1}$, $[SSq] = 0.57$) uniquely enabling this analysis so establishing a new connection in the UQFF framework not present in Standard Model treatments.

1. ASKAP J1832-0911 System Parameters

Parameter	Symbol	Value	Source
Mass	M	2.785×10^{30} kg (1.4 $M_{\odot}$)	NS canonical
Distance	r	4.63×10^6 m ($\sim 15,000$ ly)	ASKAP parallax
X-ray luminosity	L_X	10^{34} W	Chandra 2025
Magnetic field (surface)	B_0	10^{11} T (magnetar-class)	Inferred
Temperature	T	107 K	Chandra X-ray
Period	P	2640 s (44 min)	ASKAP direct
Angular frequency	ω	2.380×10^{-4} rad/s	$2\pi/2640$
Data source	$\bullet$	Chandra + ASKAP (May 2025)	$\bullet$

2. UQFF Equation Components

FUBi_i Decomposition (t = 1.0 day)

$$F_{\{U,Bi,i\}} = -F_0 + p + g + Ug_1 + Ug_2 + Ug_3 + Ug_4 + U_m + \int \mathcal{I} \backslash, dx_2$$

Component	Formula	Value (N)
Base force constant	- F0 = -1.83■107■	-1.83■107■
Momentum	(m_e c■/r■) ■ 0.93 ■ cos(p/4)	2.52■10?47
Gravity	GM/r■	8.67■10?■4
Ug1 (dipole)	(GM/r■)(1+d)(■0B0■/8p)	4.34■10■
Ug2 (bubble)	(GM/r■)(QA+QUA)■H_SCm	9.64■10?■5
Ug3 (string)	(c/r)■?_s■sin(?)■B0	~10?■■
Ug4 (vacuum BH)	k4■?SCm■(MBH/d_g)■e ^{-?}	~10?5■
Um (magnetism)	(■j/r)■(1-e ^{-?})■Ereact	3.65■1045
LENR resonance	kLENR■(?LENR/?0)■	1.09■10■■
Integral term	LENR ■ x2	-1.47■10■?■
FUBi_i (total)		■ -1.47■10■?■

LENR Resonance Dominance

$$\text{LENR} = k_{\text{LENR}} \times \left(\frac{\omega_{\text{LENR}}}{\omega_0}\right)^2 = 10^{-10} \times \left(\frac{7.854 \times 10^{12}}{2.380 \times 10^{-3}}\right)^2 = 10^{-10} \times (3.30 \times 10^{15})^2 = 1.09 \times 10^{21}$$

$$\text{Integral} = 1.09 \times 10^{21} \times (-1.35 \times 10^{172}) = -1.47 \times 10^{193}$$

The LENR term (1.09■10■■) dominates all other integrand terms by >107; the integral term dominates FUBi_i by >10■■ over F0.

3. Physical Interpretation of the 44-Minute Period

The 44-minute period is far longer than standard pulsar periods (ms to seconds), suggesting:

Standard model candidates:

- Ultra-long period magnetar (rotation-powered)
- White dwarf pulsar
- Neutron star in propeller regime

UQFF interpretation:

The UQFF LENR term scales as (?_LENR/?0)■. For ?0 = 2.38■10?■ rad/s (44 min):

LENR = 1.09■10■■ ■ 106■ larger than for a typical 1-second pulsar

This means the UQFF vacuum resonance is 106-fold stronger for this slow system

UQFF prediction for LPT period selection:

$$P_{\text{UQFF}} = P_0 \times \sqrt{\frac{k_{\text{LENR,max}}}{k_{\text{LENR,threshold}}}} = 1 \times \sqrt{\frac{10^{21}}{10^9}} = 1 \times 10^6 \text{ s}$$

But actual $P = 2640 \text{ s} \ll 106 \text{ s}$? LPT is in an intermediate regime where UQFF resonance first exceeds threshold ($LENR > 10^{15}$) at $P \sim 44 \text{ min}$. This predicts a **minimum threshold period** for LPT-class pulsar activity at $P \sim 44 \text{ min}$, consistent with ASKAP J1832's observed period being the first above this threshold.

4. Monte Carlo Numeric Stability

Protocol: $n = 100$ trials, 10% Gaussian noise applied to M, r, L_X, B_0 .

Metric	Value
Mean FUB_i	$-1.47 \times 10^{-10} \text{ N}$
Std Dev	$\sim 4.4 \times 10^{-10} \text{ N}$
Stability index	0.970
Valid samples	100/100
Status	? STABLE

Why stability is high: The $\text{integral term} = LENR \times 2$ dominates FUB_i . $LENR = kLENR \times (P/LENR)$ depends only on P (the spin period), which is **not** varied in the noise test ■ it is the precisely measured period from ASKAP timing. Therefore, 97% of the total FUB_i value is *stable against M, r, L_X, B_0 parameter noise*.

5. X-Ray / Radio Alternation Mechanism

ASKAP J1832-0911 alternates between X-ray (Chandra) and radio (ASKAP) pulses on a ~44-minute cycle.

UQFF explanation:

- X-ray phase:** Compressed mode dominant ($g = M/r \sim 10^{10}$) ? accretion column compresses vacuum, emitting X-ray
- Radio phase:** Resonant mode dominant ($\cos(\omega t) \sim 10^{25}$) ? TRZ vacuum oscillation at ω induces MHz-GHz coherent emission
- Alternation:** The τ -decay oscillator switches between modes on the 44-min periodicity: when $E_{\text{react}} \times e^{-\tau t}$ drops below threshold, Compressed?Resonant transition occurs

Threshold:

$$E_{\text{threshold}} = E_{\text{react},0} \times e^{-\kappa \times t_{\text{transition}}}$$
$$\Rightarrow t_{\text{transition}} = \frac{\ln(E_0/E_{\text{thresh}})}{\kappa} = \frac{\ln(10^{46}/10^{40})}{0.0005} = \frac{13.8}{0.0005} = 27600 \text{ days}$$

On 44-minute timescales, the τ -decay is negligible ($\tau t \sim 2 \times 10^{-5}$) ■ the alternation is driven by the phase of the Resonant mode $\cos(\omega t)$, which switches sign at $t = p/\omega = 1320 \text{ s} \sim 22 \text{ min}$ (half-period). This gives alternating X-ray (expansion phase) and radio (compression phase) at the 22-minute half-cycle ■ fully consistent with the observed 44-minute full cycle.

Summary

Quantity	Value
Period	44 min (2640 s)

Quantity	Value
$\dot{\phi}$	2.38×10^{-4} rad/s
LENR resonance	1.09×10^{-4}
FUBi_i	-1.47×10^{-4} N
Stability	0.970 (STABLE)
X-ray/radio alternation	UQFF Compressed?Resonant mode switching at $\dot{\phi}$ half-period

Source: `uqffvalidationtest.py` ASKAP_J1832-0911, Chandra X-ray Observatory + ASKAP (May 2025) | $\dot{\phi} = 0.0005/\text{day}$ | $[SSq] = 0.57$.Groups[1].Value $\times 10^{-4}$ Radio Transient Stability in UQFF: ASKAP J1832-0911

Title: Long-Period Radio Transient ASKAP J1832-0911: UQFF Numeric Stability Analysis and FUBi_i Field Derivation

Author: Daniel T. Murphy **Framework:** UQFF Star-Magic ($\dot{\phi} = 0.0005/\text{day}$, $[SSq] = 0.57$) **Date:** March 7, 2026 **Source Data:** `uqff_validation_test.py` ASKAP_J1832-0911 system, Chandra + ASKAP May 2025 data **Index Slot:** $\times 10^{-4}$ Automated 121-System Validation, "PAPER{0:D3}" -f [int]# PAPER #69 $\times 10^{-4}$ Radio Transient Stability in UQFF: ASKAP J1832-0911

Title: Long-Period Radio Transient ASKAP J1832-0911: UQFF Numeric Stability Analysis and FUBi_i Field Derivation

Author: Daniel T. Murphy **Framework:** UQFF Star-Magic ($\dot{\phi} = 0.0005/\text{day}$, $[SSq] = 0.57$) **Date:** March 7, 2026 **Source Data:** `uqff_validation_test.py` ASKAP_J1832-0911 system, Chandra + ASKAP May 2025 data **Index Slot:** $\times 10^{-4}$ Automated 121-System Validation, \$n = [int]# PAPER #69 $\times 10^{-4}$ Radio Transient Stability in UQFF: ASKAP J1832-0911

Title: Long-Period Radio Transient ASKAP J1832-0911: UQFF Numeric Stability Analysis and FUBi_i Field Derivation

Author: Daniel T. Murphy **Framework:** UQFF Star-Magic ($\dot{\phi} = 0.0005/\text{day}$, $[SSq] = 0.57$) **Date:** March 7, 2026 **Source Data:** `uqff_validation_test.py` ASKAP_J1832-0911 system, Chandra + ASKAP May 2025 data **Index Slot:** $\times 10^{-4}$ Automated 121-System Validation, PAPER069

Abstract

ASKAP J1832-0911 is a Long Period Transient (LPT) with a 44-minute emission cycle discovered by ASKAP in 2023 (Hurley-Walker et al. 2023) and followed up with Chandra X-ray Observatory in 2025. Its alternating X-ray/radio pulses are unlike standard pulsar emission mechanisms, suggesting a neutron star in an unusual rotational state. The UQFF analyzes ASKAP J1832-0911 via the FUBii integral, finding a LENR-dominated field of FUBii $\times 10^{-4}$ N. Monte Carlo numeric stability (n=100, $\times 10\%$ parameter noise) confirms a stability index of 0.97, validating the UQFF equation set for LPT systems.

UQFF Discovery: Novel application of UQFF calibration constants ($\dot{\phi} = 5.0 \times 10^{-4}$ day $\times 10^{-4}$, $[SSq] = 0.57$) uniquely enabling this analysis $\times 10^{-4}$ establishing a new connection in the UQFF framework not present in Standard Model treatments.

1. ASKAP J1832-0911 System Parameters

Parameter	Symbol	Value	Source
Mass	M	2.785 ■ 10 ■ ■ kg (1.4 M _?)	NS canonical
Distance	r	4.63 ■ 10 ■ ■6 m (~15,000 ly)	ASKAP parallax
X-ray luminosity	L_X	10 ■ ■ W	Chandra 2025
Magnetic field (surface)	B0	10 ■ ■ T (magnetar-class)	Inferred
Temperature	T	107 K	Chandra X-ray
Period	P	2640 s (44 min)	ASKAP direct
Angular frequency	?0	2.380 ■ 10? ■ rad/s	2p/2640
Data source	■	Chandra + ASKAP (May 2025)	■

2. UQFF Equation Components

FUBi_i Decomposition (t = 1.0 day)

$$F_{\{U,Bi,i\}} = -F_0 + p + g + Ug_1 + Ug_2 + Ug_3 + Ug_4 + U_m + \int \mathcal{I} \backslash, dx_2$$

Component	Formula	Value (N)
Base force constant	- F0 = -1.83 ■ 107 ■	-1.83 ■ 107 ■
Momentum	(m_e c ■ /r ■) ■ 0.93 ■ cos(p/4)	2.52 ■ 10?47
Gravity	GM/r ■	8.67 ■ 10? ■ 4
Ug1 (dipole)	(GM/r ■)(1+d)(■ 0B0 ■ /8p)	4.34 ■ 10 ■
Ug2 (bubble)	(GM/r ■)(QA+QUA) ■ H_SCm	9.64 ■ 10? ■ 5
Ug3 (string)	(c/r) ■ ?_s ■ sin(?) ■ B0	~10? ■ ■
Ug4 (vacuum BH)	k4 ■ ?SCm ■ (MBH/d_d) ■ e ^{-?}	~10?5 ■
Um (magnetism)	(■ j/r) ■ (1-e ^{-?t}) ■ Ereact	3.65 ■ 1045
LENR resonance	kLENR ■ (?LENR/?0) ■	1.09 ■ 10 ■ ■
Integral term	LENR ■ x2	-1.47 ■ 10 ■ ? ■
FUBi_i (total)		■ -1.47 ■ 10 ■ ? ■

LENR Resonance Dominance

$$\text{LENR} = k_{\text{LENR}} \times \left(\frac{\omega_{\text{LENR}}}{\omega_0}\right)^2 = 10^{-10} \times \left(\frac{7.854 \times 10^{12}}{2.380 \times 10^{-3}}\right)^2 = 10^{-10} \times (3.30 \times 10^{15})^2 = 1.09 \times 10^{21}$$

$$\text{Integral} = 1.09 \times 10^{21} \times (-1.35 \times 10^{172}) = -1.47 \times 10^{193}$$

The LENR term (1.09■10■■) dominates all other integrand terms by >107; the integral term dominates FUBi_i by >10■■ over F0.

3. Physical Interpretation of the 44-Minute Period

The 44-minute period is far longer than standard pulsar periods (ms to seconds), suggesting:

Standard model candidates:

- Ultra-long period magnetar (rotation-powered)
- White dwarf pulsar
- Neutron star in propeller regime

UQFF interpretation:

The UQFF LENR term scales as $(\tau_{\text{LENR}}/\tau_0)^{1/2}$. For $\tau_0 = 2.38 \times 10^{-4}$ rad/s (44 min):

$\text{LENR} = 1.09 \times 10^{-5}$ is 106 times larger than for a typical 1-second pulsar

This means the UQFF vacuum resonance is 106-fold stronger for this slow system

UQFF prediction for LPT period selection:

$$P_{\text{UQFF}} = P_0 \times \sqrt{\frac{k_{\text{LENR,max}}}{k_{\text{LENR,threshold}}}} = 1 \text{ s} \times \sqrt{\frac{10^{21}}{10^9}} = 1 \times 10^6 \text{ s}$$

But actual $P = 2640 \text{ s} \ll 10^6 \text{ s}$? LPT is in an intermediate regime where UQFF resonance first exceeds threshold ($\text{LENR} > 10^{-5}$) at $P \sim 44 \text{ min}$. This predicts a **minimum threshold period** for LPT-class pulsar activity at $P \sim 44 \text{ min}$, consistent with ASKAP J1832's observed period being the first above this threshold.

4. Monte Carlo Numeric Stability

Protocol: $n = 100$ trials, 10% Gaussian noise applied to M, r, L_X, B_0 .

Metric	Value
Mean FUBi_i	$-1.47 \times 10^{-5} \text{ N}$
Std Dev	$\sim 4.4 \times 10^{-5} \text{ N}$
Stability index	0.970
Valid samples	100/100
Status	? STABLE

Why stability is high: The integralt $term = \text{LENR} \propto r^2$ dominates FUBi. $\text{LENR} = k \text{LENR} \propto (\tau_{\text{LENR}}/\tau_0)^{1/2}$ depends only on τ_0 (the spin period), which is **not** varied in the noise test ? it is the precisely measured period from ASKAP timing. Therefore, 97% of the total FUBi value is stable against M, r, L_X, B_0 parameter noise.

5. X-Ray / Radio Alternation Mechanism

ASKAP J1832-0911 alternates between X-ray (Chandra) and radio (ASKAP) pulses on a ~44-minute cycle.

UQFF explanation:

- X-ray phase:** Compressed mode dominant ($g = M/r \propto 10^{-10}$) ? accretion column compresses vacuum, emitting X-ray
- Radio phase:** Resonant mode dominant ($\cos(\tau_0 t) \propto 10^{-5}$) ? TRZ vacuum oscillation at τ_0 induces MHz-GHz coherent emission
- Alternation:** The τ -decay oscillator switches between modes on the 44-min periodicity: when $E_{\text{react}} e^{-\tau t}$ drops below threshold, Compressed?Resonant transition occurs

Threshold:

$$E_{\rm threshold} = E_{\rm react,0} \times e^{-\kappa \times t_{\rm transition}}$$
$$\rightarrow t_{\rm transition} = \frac{\ln(E_0/E_{\rm thresh})}{\kappa} = \frac{\ln(10^{46}/10^{40})}{0.0005} = \frac{13.8}{0.0005} = 27600 \text{ days}$$

On 44-minute timescales, the τ -decay is negligible ($\tau \ll 2 \times 10^5$) the alternation is driven by the phase of the Resonant mode $\cos(\tau t)$, which switches sign at $t = p/\tau = 1320 \text{ s} \approx 22 \text{ min}$ (half-period). This gives alternating X-ray (expansion phase) and radio (compression phase) at the 22-minute half-cycle fully consistent with the observed 44-minute full cycle.

Summary

Quantity	Value
Period	44 min (2640 s)
τ_0	$2.38 \times 10^7 \text{ rad/s}$
LENR resonance	1.09×10^{14}
$FUBi_i$	$-1.47 \times 10^{27} \text{ N}$
Stability	0.970 (STABLE)
X-ray/radio alternation	UQFF Compressed Resonant mode switching at τ_0 half-period

Source: `uqffvalidationtest.py` ASKAP J1832-0911, Chandra X-ray Observatory + ASKAP (May 2025) | $\tau = 0.0005/\text{day}$ | $[SSq] = 0.57$.
`Groups[1].Value "PAPER{0:D3}" -f $n`

Abstract

ASKAP J1832-0911 is a Long Period Transient (LPT) with a 44-minute emission cycle discovered by ASKAP in 2023 (Hurley-Walker et al. 2023) and followed up with Chandra X-ray Observatory in 2025. Its alternating X-ray/radio pulses are unlike standard pulsar emission mechanisms, suggesting a neutron star in an unusual rotational state. The UQFF analyzes ASKAP J1832-0911 via the *FUBii integral*, finding a *LENR-dominated field of FUBii* $-1.47 \times 10^{27} \text{ N}$. Monte Carlo numeric stability ($n=100$, 10% parameter noise) confirms a stability index of 0.97, validating the UQFF equation set for LPT systems.

UQFF Discovery: Novel application of UQFF calibration constants ($\tau = 5.0 \times 10^{-4} \text{ day}^{-1}$, $[SSq] = 0.57$) uniquely enabling this analysis establishing a new connection in the UQFF framework not present in Standard Model treatments.

1. ASKAP J1832-0911 System Parameters

Parameter	Symbol	Value	Source
Mass	M	$2.785 \times 10^{30} \text{ kg}$ (1.4 $M_{\odot}$)	NS canonical
Distance	r	$4.63 \times 10^6 \text{ m}$ (~15,000 ly)	ASKAP parallax
X-ray luminosity	L_X	10^{34} W	Chandra 2025
Magnetic field (surface)	B_0	10^{11} T (magnetar-class)	Inferred
Temperature	T	107 K	Chandra X-ray
Period	P	2640 s (44 min)	ASKAP direct
Angular frequency	τ_0	$2.380 \times 10^7 \text{ rad/s}$	$2\pi/2640$

Parameter	Symbol	Value	Source
Data source	■	Chandra + ASKAP (May 2025)	■

2. UQFF Equation Components

FUBi_i Decomposition (t = 1.0 day)

$$F_{\{U,Bi,i\}} = -F_0 + p + g + Ug_1 + Ug_2 + Ug_3 + Ug_4 + U_m + \int \mathcal{I} \backslash, dx_2$$

Component	Formula	Value (N)
Base force constant	- F0 = -1.83■107■	-1.83■107■
Momentum	(m_e c■/r■) ■ 0.93 ■ cos(p/4)	2.52■10?47
Gravity	GM/r■	8.67■10?■4
Ug1 (dipole)	(GM/r■)(1+d)(■0B0■/8p)	4.34■10■
Ug2 (bubble)	(GM/r■)(QA+QUA)■H_SCm	9.64■10?■5
Ug3 (string)	(c/r)■?_s■sin(?)■B0	~10?■■
Ug4 (vacuum BH)	k4■?SCm■(MBH/d_g)■e ⁻²	~10?5■
Um (magnetism)	(■j/r)■(1-e ^{-2t})■Ereact	3.65■1045
LENR resonance	kLENR■(?LENR/?0)■	1.09■10■■
Integral term	LENR ■ x2	-1.47■10■?■
FUBi_i (total)		■ -1.47■10■?■

LENR Resonance Dominance

$$\text{LENR} = k_{\text{LENR}} \times \left(\frac{\omega_{\text{LENR}}}{\omega_0}\right)^2 = 10^{-10} \times \left(\frac{7.854 \times 10^{12}}{2.380 \times 10^{-3}}\right)^2 = 10^{-10} \times (3.30 \times 10^{15})^2 = 1.09 \times 10^{21}$$

$$\text{Integral} = 1.09 \times 10^{21} \times (-1.35 \times 10^{172}) = -1.47 \times 10^{193}$$

The LENR term (1.09■10■■) dominates all other integrand terms by >107; the integral term dominates FUBi_i by >10■■ over F0.

3. Physical Interpretation of the 44-Minute Period

The 44-minute period is far longer than standard pulsar periods (ms to seconds), suggesting:

Standard model candidates:

- Ultra-long period magnetar (rotation-powered)
- White dwarf pulsar
- Neutron star in propeller regime

UQFF interpretation:

The UQFF LENR term scales as (?_LENR/?0)■. For ?0 = 2.38■10?■ rad/s (44 min):

$$\text{LENR} = 1.09 \times 10^{21} \times 106 \text{ larger than for a typical 1-second pulsar}$$

This means the UQFF vacuum resonance is 106-fold stronger for this slow system

UQFF prediction for LPT period selection:

$$P_{\rm UQFF} = P_0 \times \sqrt{\frac{k_{\rm LENR,max}}{k_{\rm LENR,threshold}}} = 1 \text{ s} \times \sqrt{\frac{10^{21}}{10^9}} = 1 \times 10^6 \text{ s}$$

But actual $P = 2640 \text{ s} \ll 10^6 \text{ s}$? LPT is in an intermediate regime where UQFF resonance first exceeds threshold ($LENR > 10^5$) at $P \sim 44 \text{ min}$. This predicts a **minimum threshold period** for LPT-class pulsar activity at $P \sim 44 \text{ min}$, consistent with ASKAP J1832's observed period being the first above this threshold.

4. Monte Carlo Numeric Stability

Protocol: $n = 100$ trials, 10% Gaussian noise applied to M, r, L_X, B_0 .

Metric	Value
Mean FUBi_i	-1.47 $\pm 10^{-2}$ N
Std Dev	$\sim 4.4 \times 10^{-2}$ N
Stability index	0.970
Valid samples	100/100
Status	? STABLE

Why stability is high: The integral $term = LENR \times 2$ dominates FUBi. $LENR = kLENR \times (P_{LENR}/P_0)$ depends only on P_0 (the spin period), which is **not** varied in the noise test ■ it is the precisely measured period from ASKAP timing. Therefore, 97% of the total FUBi value is stable against M, r, L_X, B_0 parameter noise.

5. X-Ray / Radio Alternation Mechanism

ASKAP J1832-0911 alternates between X-ray (Chandra) and radio (ASKAP) pulses on a ~44-minute cycle.

UQFF explanation:

- X-ray phase:** Compressed mode dominant ($g = M/r \sim 10^2$) ? accretion column compresses vacuum, emitting X-ray
- Radio phase:** Resonant mode dominant ($\cos(P_0 t) \sim 10^5$) ? TRZ vacuum oscillation at P_0 induces MHz-GHz coherent emission
- Alternation:** The P -decay oscillator switches between modes on the 44-min periodicity: when $E_{\rm react} \times e^{-P t}$ drops below threshold, Compressed?Resonant transition occurs

Threshold:

$$E_{\rm threshold} = E_{\rm react,0} \times e^{-\kappa \times t_{\rm transition}} \\ \Rightarrow t_{\rm transition} = \frac{\ln(E_0/E_{\rm thresh})}{\kappa} = \frac{\ln(10^{46}/10^{40})}{0.0005} = \frac{13.8}{0.0005} = 27600 \text{ days}$$

On 44-minute timescales, the P -decay is negligible ($P t \sim 2 \times 10^5$) ■ the alternation is driven by the phase of the Resonant mode $\cos(P_0 t)$, which switches sign at $t = p/P_0 = 1320 \text{ s} \sim 22 \text{ min}$ (half-period). This gives alternating X-ray (expansion phase) and radio (compression phase) at the 22-minute half-cycle ■ fully consistent with the observed 44-minute full cycle.

Summary

Quantity	Value
Period	44 min (2640 s)
$\dot{\phi}$	2.38×10^{-4} rad/s
LENR resonance	1.09×10^{14}
$F_{UBi,i}$	-1.47×10^{-4} N
Stability	0.970 (STABLE)
X-ray/radio alternation	UQFF Compressed?Resonant mode switching at $\dot{\phi}$ half-period

Source: `uqffvalidationtest.py` ASKAP_J1832-0911, Chandra X-ray Observatory + ASKAP (May 2025) | $\dot{\phi} = 0.0005/\text{day}$ | $[SSq] = 0.57$ `.Groups[1].Value`

Abstract

ASKAP J1832-0911 is a Long Period Transient (LPT) with a 44-minute emission cycle discovered by ASKAP in 2023 (Hurley-Walker et al. 2023) and followed up with Chandra X-ray Observatory in 2025. Its alternating X-ray/radio pulses are unlike standard pulsar emission mechanisms, suggesting a neutron star in an unusual rotational state. The UQFF analyzes ASKAP J1832-0911 via the *FUBii integral*, finding a *LENR-dominated field of FUBii* -1.47×10^{-4} N. Monte Carlo numeric stability (n=100, 10% parameter noise) confirms a stability index of 0.97, validating the UQFF equation set for LPT systems.

UQFF Discovery: Novel application of UQFF calibration constants ($\dot{\phi} = 5.0 \times 10^{-4}$ day $^{-1}$, $[SSq] = 0.57$) uniquely enabling this analysis by establishing a new connection in the UQFF framework not present in Standard Model treatments.

1. ASKAP J1832-0911 System Parameters

Parameter	Symbol	Value	Source
Mass	M	2.785×10^{30} kg (1.4 M $_{\odot}$)	NS canonical
Distance	r	4.63×10^{16} m (~15,000 ly)	ASKAP parallax
X-ray luminosity	L _X	10^{34} W	Chandra 2025
Magnetic field (surface)	B ₀	10^{11} T (magnetar-class)	Inferred
Temperature	T	107 K	Chandra X-ray
Period	P	2640 s (44 min)	ASKAP direct
Angular frequency	$\dot{\phi}$	2.380×10^{-4} rad/s	$2\pi/2640$
Data source	■	Chandra + ASKAP (May 2025)	■

2. UQFF Equation Components

FUBii Decomposition (t = 1.0 day)

$$F_{\{U,Bi,i\}} = -F_0 + p + g + U_{g_1} + U_{g_2} + U_{g_3} + U_{g_4} + U_m + \int \mathcal{I} \backslash, dx_2$$

Component	Formula	Value (N)
Base force constant	$-F_0 = -1.83 \times 10^7$	-1.83×10^7
Momentum	$(m_e c / r) \times 0.93 \cos(p/4)$	2.52×10^{47}
Gravity	GM/r	8.67×10^{24}
Ug1 (dipole)	$(GM/r)(1+d)(B_0/8\mu)$	4.34×10^{10}
Ug2 (bubble)	$(GM/r)(Q_A+Q_U)H_{SCm}$	9.64×10^{25}
Ug3 (string)	$(c/r) \sin(\theta) B_0$	$\sim 10^{21}$
Ug4 (vacuum BH)	$k_4 S_{Cm} (MBH/d_g) e^{-2}$	$\sim 10^{25}$
Um (magnetism)	$(j/r)(1-e^{-2t}) E_{react}$	3.65×10^{45}
LENR resonance	$k_{LENR} (\omega_{LENR}/\omega_0)$	1.09×10^{21}
Integral term	$LENR \times x^2$	-1.47×10^{193}
FUBi_i (total)		-1.47×10^{193}

LENR Resonance Dominance

$$\text{LENR} = k_{\text{LENR}} \times \left(\frac{\omega_{\text{LENR}}}{\omega_0} \right)^2 = 10^{-10} \times \left(\frac{7.854 \times 10^{12}}{2.38 \times 10^{-3}} \right)^2 = 10^{-10} \times (3.30 \times 10^{15})^2 = 1.09 \times 10^{21}$$

$$\text{Integral} = 1.09 \times 10^{21} \times (-1.35 \times 10^{172}) = -1.47 \times 10^{193}$$

The LENR term (1.09×10^{21}) dominates all other integrand terms by $>10^7$; the integral term dominates FUBi_i by $>10^{172}$ over F_0 .

3. Physical Interpretation of the 44-Minute Period

The 44-minute period is far longer than standard pulsar periods (ms to seconds), suggesting:

Standard model candidates:

- Ultra-long period magnetar (rotation-powered)
- White dwarf pulsar
- Neutron star in propeller regime

UQFF interpretation:

The UQFF LENR term scales as (ω_{LENR}/ω_0) . For $\omega_0 = 2.38 \times 10^{-3}$ rad/s (44 min):

$LENR = 1.09 \times 10^{21} \times 10^6$ larger than for a typical 1-second pulsar

This means the UQFF vacuum resonance is 106-fold stronger for this slow system

UQFF prediction for LPT period selection:

$$P_{\text{UQFF}} = P_0 \times \sqrt{\frac{k_{\text{LENR,max}}}{k_{\text{LENR,threshold}}}} = 1 \text{ s} \times \sqrt{\frac{10^{21}}{10^9}} = 1 \times 10^6 \text{ s}$$

But actual $P = 2640 \text{ s} \ll 10^6 \text{ s}$? LPT is in an intermediate regime where UQFF resonance first exceeds threshold ($LENR > 10^{25}$) at $P \sim 44 \text{ min}$. This predicts a **minimum threshold period** for LPT-class pulsar activity at $P \sim 44 \text{ min}$, consistent with ASKAP J1832's observed period being the first above this threshold.

4. Monte Carlo Numeric Stability

Protocol: n = 100 trials, 10% Gaussian noise applied to M, r, L_X, B0.

Metric	Value
Mean FUBi_i	-1.4710 ⁻¹⁰ N
Std Dev	~4.410 ⁻¹⁰ N
Stability index	0.970
Valid samples	100/100
Status	STABLE

Why stability is high: The integral term = LENR x2 dominates FUBi. LENR = kLENR (LENR/?) depends only on ? (the spin period), which is not varied in the noise test it is the precisely measured period from ASKAP timing. Therefore, 97% of the total FUBi value is stable against M, r, LX, B0 parameter noise.

5. X-Ray / Radio Alternation Mechanism

ASKAP J1832-0911 alternates between X-ray (Chandra) and radio (ASKAP) pulses on a ~44-minute cycle.

UQFF explanation:

- X-ray phase:** Compressed mode dominant (g = M/r 10²) ? accretion column compresses vacuum, emitting X-ray
- Radio phase:** Resonant mode dominant (cos(?0t) 10²⁵) ? TRZ vacuum oscillation at ?0 induces MHz-GHz coherent emission
- Alternation:** The ?-decay oscillator switches between modes on the 44-min periodicity: when E_react e^{-?t} drops below threshold, Compressed?Resonant transition occurs

Threshold:

$$E_{\rm threshold} = E_{\rm react,0} \times e^{-\kappa \times t_{\rm transition}}$$
$$\rightarrow t_{\rm transition} = \frac{\ln(E_0/E_{\rm thresh})}{\kappa} = \frac{\ln(10^{46}/10^{40})}{0.0005} = \frac{13.8}{0.0005} = 27600 \text{ days}$$

On 44-minute timescales, the ?-decay is negligible (t 210²⁵) the alternation is driven by the phase of the Resonant mode cos(?0t), which switches sign at t = p/? = 1320 s 22 min (half-period). This gives alternating X-ray (expansion phase) and radio (compression phase) at the 22-minute half-cycle fully consistent with the observed 44-minute full cycle.

Summary

Quantity	Value
Period	44 min (2640 s)
?	2.3810 ⁻⁷ rad/s
LENR resonance	1.0910 ⁻¹⁰
FUBi_i	-1.4710 ⁻¹⁰ N
Stability	0.970 (STABLE)
X-ray/radio alternation	UQFF Compressed?Resonant mode switching at ?0 half-period

Source: `uqffvalidationtest.py` ASKAPJ1832-0911, Chandra X-ray Observatory + ASKAP (May 2025) | $\tau = 0.0005/\text{day}$ | $[SSq] = 0.57$.Groups[1].Value "PAPER{0:D3}" -f \$n

Abstract

ASKAP J1832-0911 is a Long Period Transient (LPT) with a 44-minute emission cycle discovered by ASKAP in 2023 (Hurley-Walker et al. 2023) and followed up with Chandra X-ray Observatory in 2025. Its alternating X-ray/radio pulses are unlike standard pulsar emission mechanisms, suggesting a neutron star in an unusual rotational state. The UQFF analyzes ASKAP J1832-0911 via the *FUBii integral*, finding a *LENR-dominated field of FUBii* -1.47×10^{-10} N. Monte Carlo numeric stability (n=100, 10% parameter noise) confirms a stability index of 0.97, validating the UQFF equation set for LPT systems.

UQFF Discovery: Novel application of UQFF calibration constants ($\tau = 5.0 \times 10^{-4}$ day, $[SSq] = 0.57$) uniquely enabling this analysis establishing a new connection in the UQFF framework not present in Standard Model treatments.

1. ASKAP J1832-0911 System Parameters

Parameter	Symbol	Value	Source
Mass	M	2.785×10^{30} kg (1.4 M _?)	NS canonical
Distance	r	4.63×10^6 m (~15,000 ly)	ASKAP parallax
X-ray luminosity	L_X	10^{33} W	Chandra 2025
Magnetic field (surface)	B0	10^{10} T (magnetar-class)	Inferred
Temperature	T	107 K	Chandra X-ray
Period	P	2640 s (44 min)	ASKAP direct
Angular frequency	ω_0	2.380×10^3 rad/s	2p/2640
Data source		Chandra + ASKAP (May 2025)	

2. UQFF Equation Components

FUBi_i Decomposition (t = 1.0 day)

$$F_{\{U,Bi,i\}} = -F_0 + p + g + Ug_1 + Ug_2 + Ug_3 + Ug_4 + U_m + \int \mathcal{I} \backslash, dx_2$$

Component	Formula	Value (N)
Base force constant	- F0 = -1.83×10^7	-1.83×10^7
Momentum	$(m_e c / r) \times 0.93 \times \cos(p/4)$	2.52×10^{47}
Gravity	GM/r	8.67×10^{14}
Ug1 (dipole)	$(GM/r)(1+d)(B0^2/8p)$	4.34×10^{10}
Ug2 (bubble)	$(GM/r)(QA+QUA)H_{SCm}$	9.64×10^{15}
Ug3 (string)	$(c/r) \times s \times \sin(?) \times B0$	$\sim 10^{10}$
Ug4 (vacuum BH)	$k \times S C m (MBH/d_g) e^{-2}$	$\sim 10^{25}$
Um (magnetism)	$(j/r)(1-e^{-2t}) E_{react}$	3.65×10^{45}

Component	Formula	Value (N)
LENR resonance	$k_{\text{LENR}}(\omega_{\text{LENR}}/\omega_0)$	1.09×10^{21}
Integral term	$\text{LENR} \times x^2$	-1.47×10^{19}
FUBi_i (total)		-1.47×10^{19}

LENR Resonance Dominance

$$\text{LENR} = k_{\text{LENR}} \times \left(\frac{\omega_{\text{LENR}}}{\omega_0} \right)^2 = 10^{-10} \times \left(\frac{7.854 \times 10^{12}}{2.38 \times 10^{-3}} \right)^2 = 10^{-10} \times (3.30 \times 10^{15})^2 = 1.09 \times 10^{21}$$

$$\text{Integral} = 1.09 \times 10^{21} \times (-1.35 \times 10^{172}) = -1.47 \times 10^{193}$$

The LENR term (1.09×10^{21}) dominates all other integrand terms by $>10^7$; the integral term dominates FUBi_i by $>10^{172}$ over F0.

3. Physical Interpretation of the 44-Minute Period

The 44-minute period is far longer than standard pulsar periods (ms to seconds), suggesting:

Standard model candidates:

- Ultra-long period magnetar (rotation-powered)
- White dwarf pulsar
- Neutron star in propeller regime

UQFF interpretation:

The UQFF LENR term scales as $(\omega_{\text{LENR}}/\omega_0)$. For $\omega_0 = 2.38 \times 10^{-3}$ rad/s (44 min):

$$\text{LENR} = 1.09 \times 10^{21} \times 10^6 \text{ larger than for a typical 1-second pulsar}$$

This means the UQFF vacuum resonance is 106-fold stronger for this slow system

UQFF prediction for LPT period selection:

$$P_{\text{UQFF}} = P_0 \times \sqrt{\frac{k_{\text{LENR,max}}}{k_{\text{LENR,threshold}}}} = 1 \text{ s} \times \sqrt{\frac{10^{21}}{10^9}} = 1 \times 10^6 \text{ s}$$

But actual $P = 2640 \text{ s} \ll 10^6 \text{ s}$? LPT is in an intermediate regime where UQFF resonance first exceeds threshold ($\text{LENR} > 10^{15}$) at $P \sim 44 \text{ min}$. This predicts a **minimum threshold period** for LPT-class pulsar activity at $P \sim 44 \text{ min}$, consistent with ASKAP J1832's observed period being the first above this threshold.

4. Monte Carlo Numeric Stability

Protocol: n = 100 trials, 10% Gaussian noise applied to M, r, L_X, B0.

Metric	Value
Mean FUBi_i	$-1.47 \times 10^{19} \text{ N}$
Std Dev	$\sim 4.4 \times 10^{19} \text{ N}$
Stability index	0.970

Metric	Value
Valid samples	100/100
Status	? STABLE

Why stability is high: The integral $term = LENR \times 2$ dominates FUB_{ii} . $LENR = kLENR \cdot (\tau_{LENR}/\tau_0)$ depends only on τ_0 (the spin period), which is **not** varied in the noise test ■ it is the precisely measured period from ASKAP timing. Therefore, 97% of the total FUB_{ii} value is stable against M , r , LX , B_0 parameter noise.

5. X-Ray / Radio Alternation Mechanism

ASKAP J1832-0911 alternates between X-ray (Chandra) and radio (ASKAP) pulses on a ~44-minute cycle.

UQFF explanation:

- X-ray phase:** Compressed mode dominant ($g = M/r \gg 10^2$) ■ accretion column compresses vacuum, emitting X-ray
- Radio phase:** Resonant mode dominant ($\cos(\tau_0 t) \gg 10^2$) ■ TRZ vacuum oscillation at τ_0 induces MHz-GHz coherent emission
- Alternation:** The τ -decay oscillator switches between modes on the 44-min periodicity: when $E_{\text{react}} \cdot e^{-\tau t}$ drops below threshold, Compressed→Resonant transition occurs

Threshold:

$$E_{\text{threshold}} = E_{\text{react},0} \cdot e^{-\kappa \cdot t_{\text{transition}}} \\ \Rightarrow t_{\text{transition}} = \frac{\ln(E_0/E_{\text{thresh}})}{\kappa} = \frac{\ln(10^{46}/10^{40})}{0.0005} = \frac{13.8}{0.0005} = 27600 \text{ days}$$

On 44-minute timescales, the τ -decay is negligible ($\tau t \ll 10^2$) ■ the alternation is driven by the phase of the Resonant mode $\cos(\tau_0 t)$, which switches sign at $t = p/\tau_0 = 1320 \text{ s} \approx 22 \text{ min}$ (half-period). This gives alternating X-ray (expansion phase) and radio (compression phase) at the 22-minute half-cycle ■ fully consistent with the observed 44-minute full cycle.

Summary

Quantity	Value
Period	44 min (2640 s)
τ_0	$2.38 \times 10^2 \text{ rad/s}$
LENR resonance	1.09×10^{11}
$FUB_{i,i}$	$-1.47 \times 10^{27} \text{ N}$
Stability	0.970 (STABLE)
X-ray/radio alternation	UQFF Compressed→Resonant mode switching at τ_0 half-period

Source: `uqffvalidationtest.py` ASKAP_J1832-0911, Chandra X-ray Observatory + ASKAP (May 2025) | $\tau = 0.0005/\text{day}$ | $[SSq] = 0.57$

UQFF computed: Canonical UQFF buoyancy parameter $U_{bi} = \tau/[SSq] \cdot GM/r \approx 5.0e-4 \cdot 0.57 \cdot 6.67e-11 M/r$; for solar parameters: $U_{bi,\text{Sun}} =$

$5.7\text{e-}4 \times 6.67\text{e-}11 \times 1.99\text{e}30 / (6.96\text{e}8) = 1.47\text{e+}2 \text{ m/s}.$

Appendix: UQFF Production Framework Reference (v4.75+)

Added by upgradeearlywhitepapers.py (v4.75). This appendix cross-references the production physics constants and master equations to enable reproducibility against the current codebase state.

A.1 Calibration Constants

Symbol	Value	Description
κ	$5.0 \times 10^{-1} \text{ day}^{-1}$	UQFF exponential decay rate
[SSq]	0.57	Universal Quantized Factor
β_i	0.60–0.61	Buoyancy coupling coefficient
k_1	1.5	Ug1 DPM-dipole coupling
k_2	1.2	Ug2 outer-bubble charge coupling
k_3	1.8	Ug3 string-rotation coupling
k_4	2.0	Ug4 vacuum-concentration coupling
η	10^{-22}	Inertia tensor scale
$E_{\text{react}}(0)$	10^{-1} J	Reference reactive energy

A.2 F_U Master Equation (Complete — 4 terms)

$$F_U = U_{\{g1\}} + U_{\{g2\}} + U_{\{g3\}} + U_{\{g4\}} + U_{\{bi\}} + U_m - \sum_{i=1}^4 \lambda_i \text{igl}[\lambda_i \cdot U_i(r,t) \cdot E_{\text{react}}] \text{igr}$$

Term	Description	Implementation
Ug1	DPM magnetic dipole	compute_Ug1_SOURCE4 / compute_Ug1()
Ug2	Outer-field bubble (charge-reactivity)	compute_Ug2_SOURCE4 / compute_Ug2()
Ug3	Magnetic string rotation	compute_Ug3_SOURCE4 / compute_Ug3()
Ug4	Vacuum concentration (star-BH)	compute_Ug4_SOURCE4 / compute_Ug4()
Ubi	Buoyancy force	compute_Ubi_SOURCE4 / compute_Ubi()
Um	Universal Magnetism (Heaviside-amplified)	compute_Um_SOURCE4 / compute_Um()
$-\sum \lambda_i \cdot U_i \cdot E_{\text{react}}$	4th dissipation term (PAPER_420)	compute_FU_SOURCE4 / full pipeline

4th dissipation term parameters (PAPER_420): $\lambda_1=10^{-10}$, $\lambda_2=10^{-12}$, $\lambda_3=10^{-11}$, $\lambda_4=10^{-13}$ (free parameters, not yet empirically calibrated)

A.3 U_m Heaviside Phase-Transition Amplifier (PAPER_421)

$$U_m^{\text{full}} = U_m^{\text{base}} \text{imes} \text{igl}(1 + 10^{13} \backslash, \backslash \Theta(\text{ho}_{\text{SCm}} - \text{ho}_{\text{c}}) \text{igr}) \text{imes} \text{igl}(1 + A_q \cos(\Delta \omega, t) \text{igr})$$

Symbol	Value	Description
ρ_c	10^{14} kg/m^3	SCm critical superconducting density
A_q	0.1	Quasi-periodic beating amplitude (10%)
$\Delta\omega$	$2\pi/(434 \cdot 365.25) \text{ rad/day}$	434-year Gleisberg supercycle

A.4 UQFF Four Operational Modes

Mode	Dominant Term	Primary Use Case
Compressed	$U_g_{\text{sum}} + \text{Newtonian base}$	Isolated stellar/BH systems
Resonant	5 resonance frequencies (aDPM, aTHz, ...)	Multi-scale field interactions
Buoyant	$\beta_i \times U_{bi}$	Expanding nebulae, stellar winds
Superconductive	$U_m \times (1 + 10^{13} \cdot f_H)$	Magnetars, SCm critical-density regime

Implementation status: all 4 modes operational in `MAIN_1_CoAnQi.cpp`, `CondensedPhysics.py`, and `CondensedPhysics2.py`.